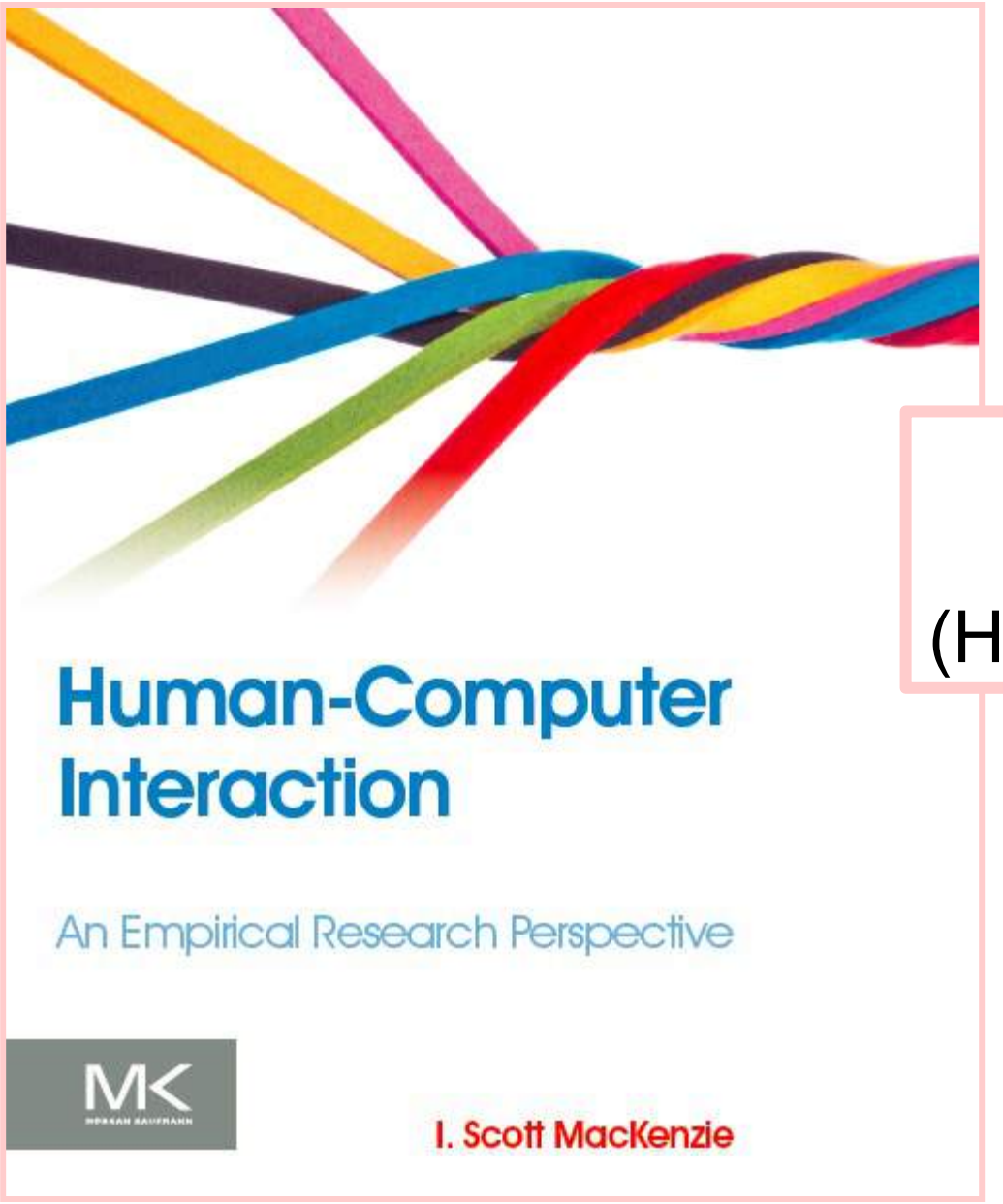
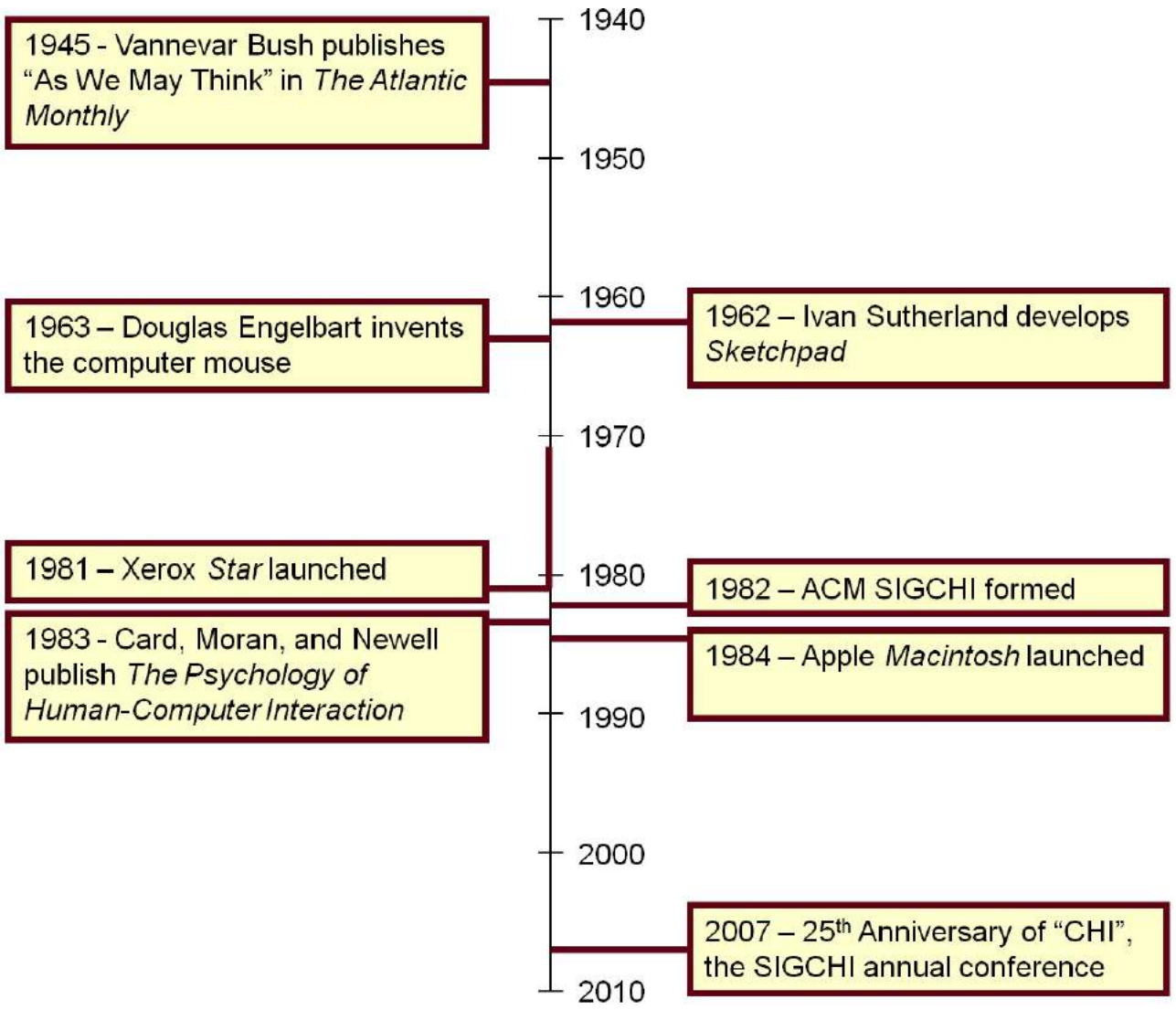


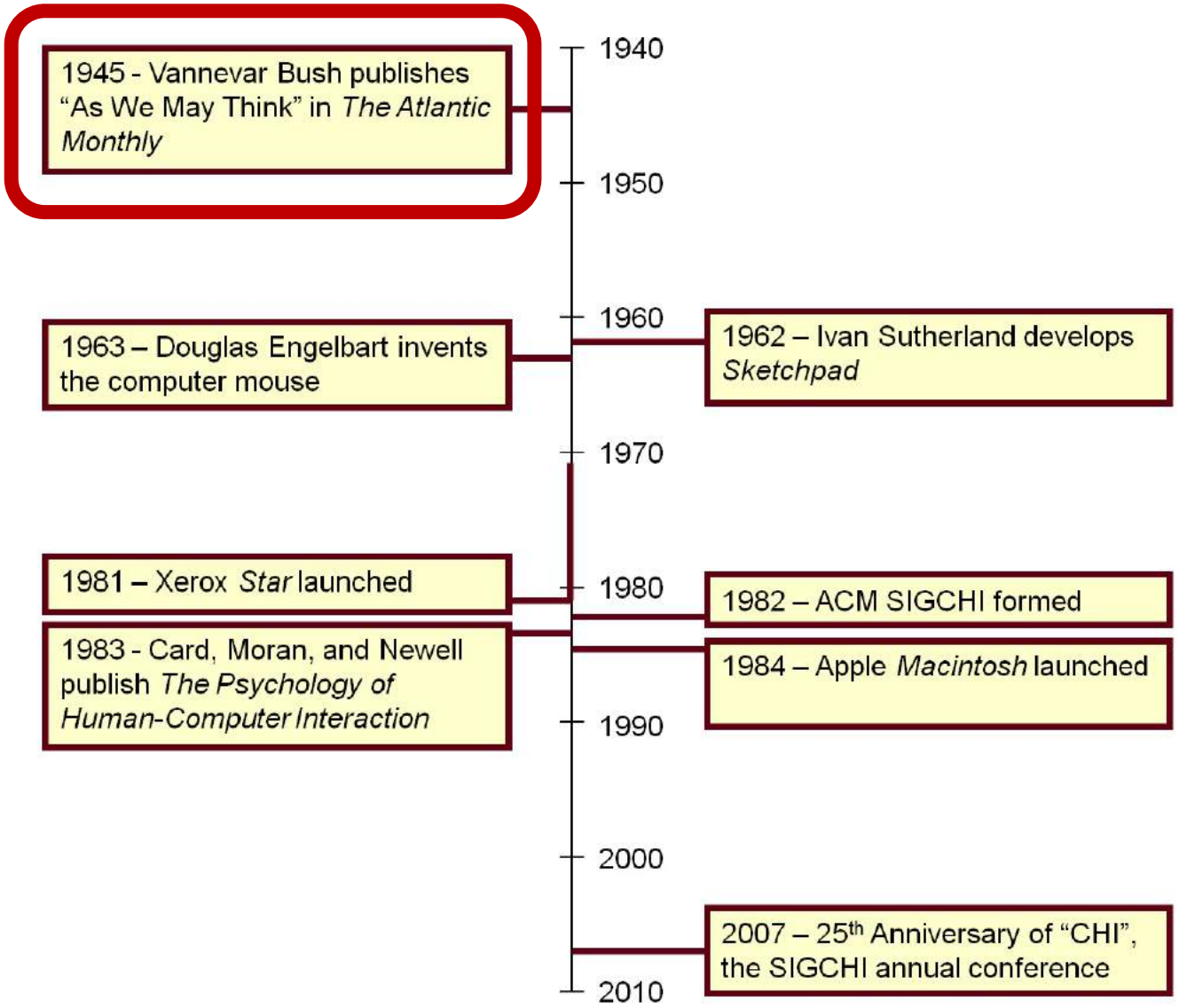
Chapter 1 Introduction (Historical Perspective)

Longbiao CHEN
2018/04 Xiamen University

Timeline of Significant Events



Timeline of Significant Events



“As We May Think”
Vannevar Bush (1945)



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As we may think

Full Text: Pdf

Author: Vannevar Bush Director of the Office of Scientific Research and Development

Published in:

· Magazine
interactions Interactions Homepage archive
Volume 3 Issue 2, March 1996
Pages 35 - 46
ACM New York, NY, USA
table of contents doi>10.1145/227181.227186



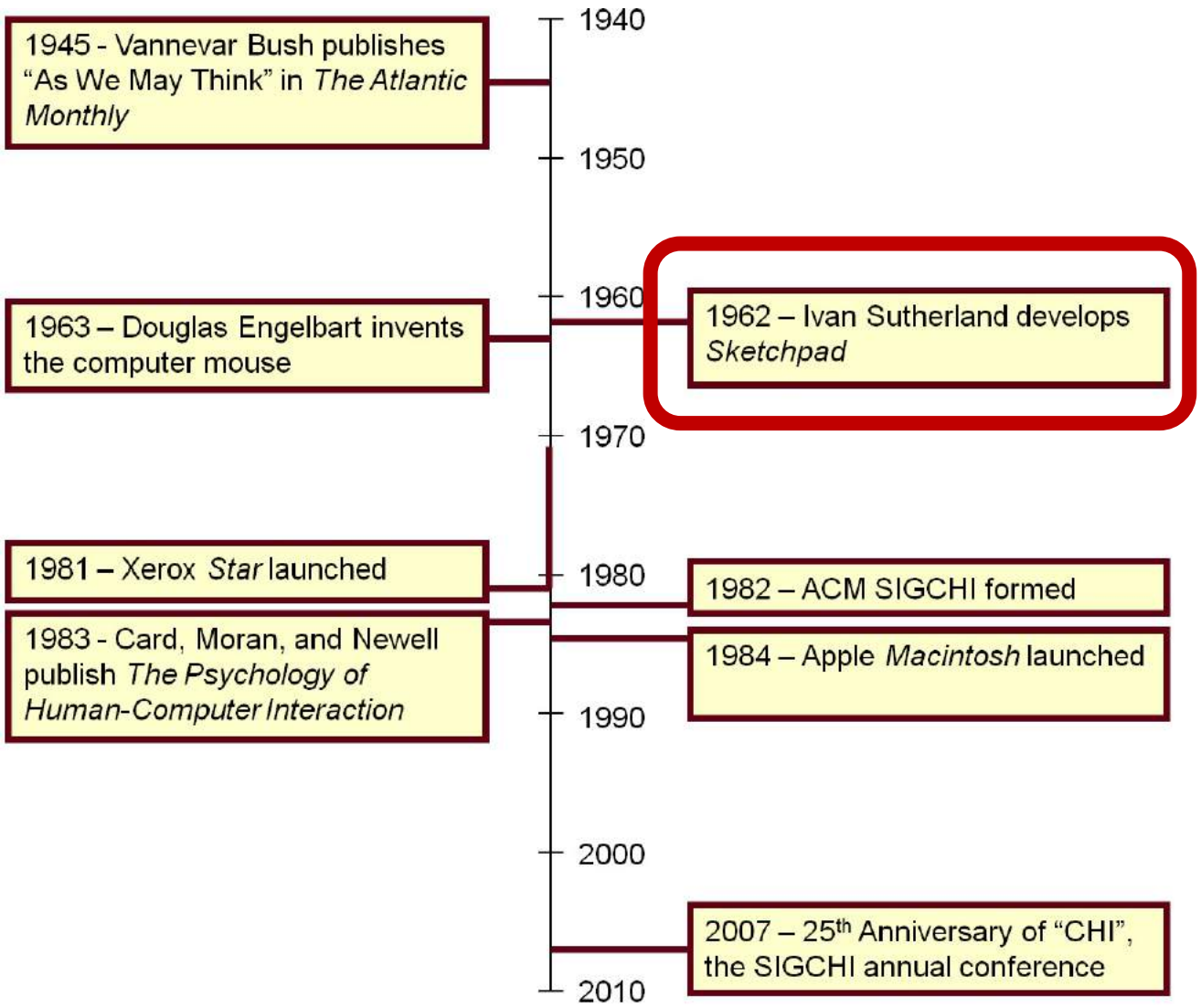
1996 Article

Bibliometrics
- Downloads (6 Weeks): 54
- Downloads (12 Months): 446
- Citation Count: 19

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http://dl.acm.org/citation.cfm?id=227186

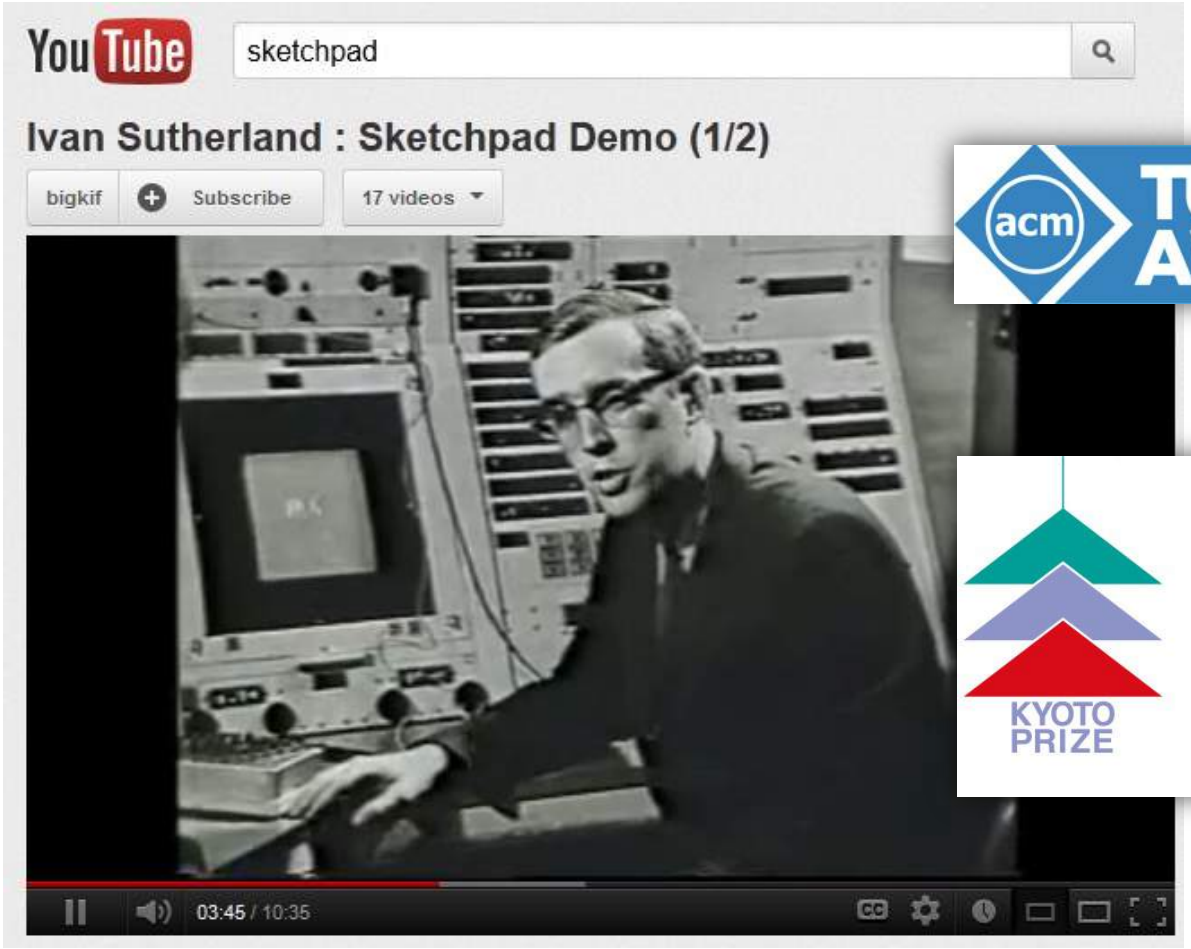
Timeline of Significant Events



Sketchpad
Ivan Sutherland (1962)



Viewable on…



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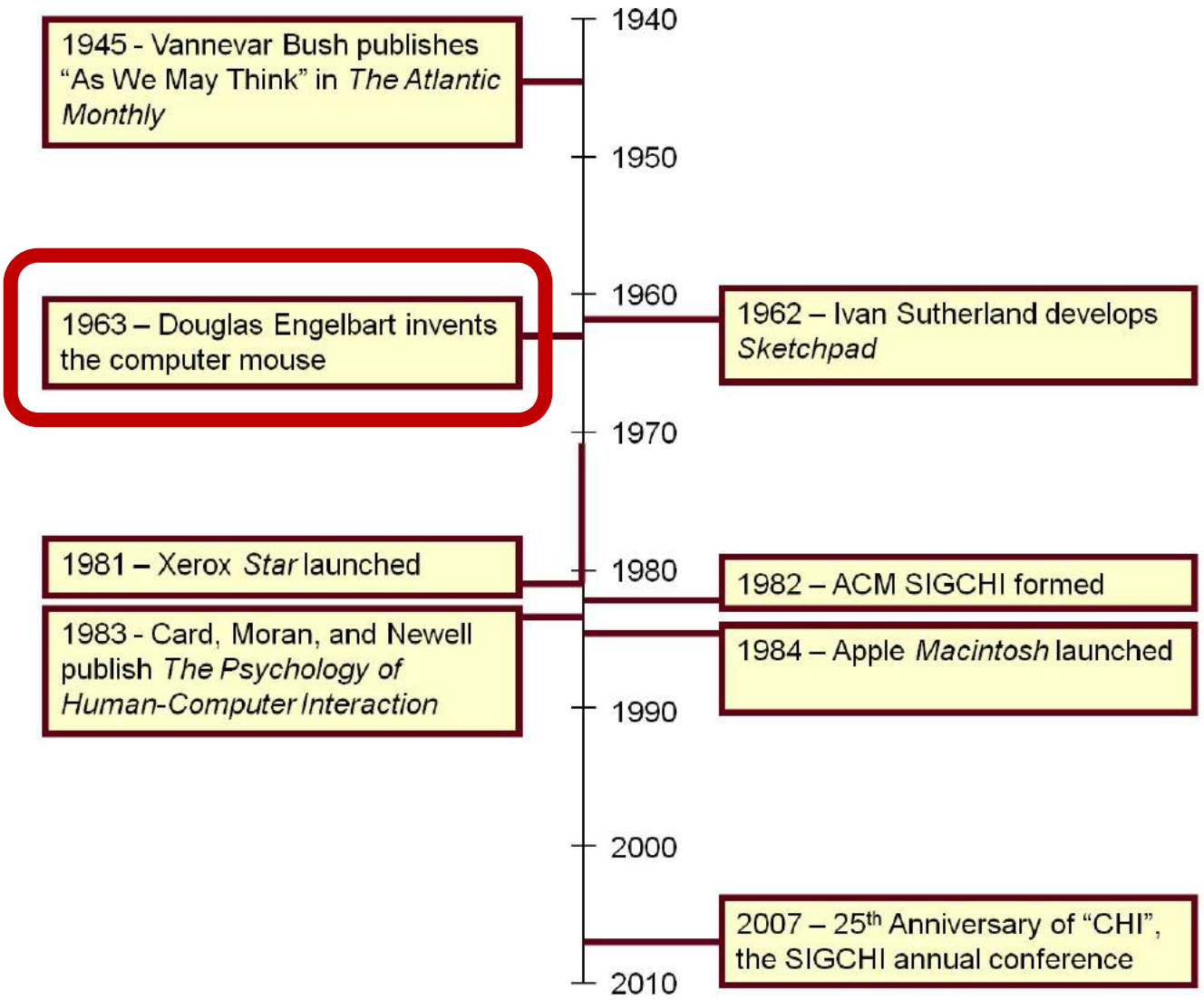
https://www.youtube.com/watch?v=USyoT_Ha_bA

Sketchpad: "Direct Manipulation"

- Direct manipulation features:
  - Visibility of objects
  - Incremental action and rapid feedback
  - Reversibility
  - Exploration
  - Syntactic correctness of all actions
  - Replacing language with action
- Term coined by Ben Shneiderman[1]

[1] Shneiderman, B., Direct manipulation: A step beyond programming languages, in *IEEE Computer*, 1983, August, 57-69.

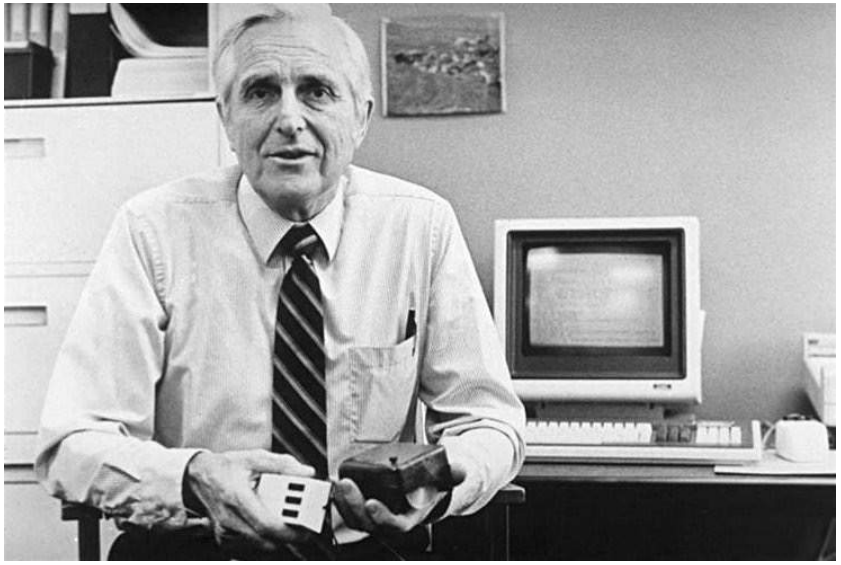
Timeline of Significant Events



Invention of the Mouse

Doug Engelbart (1963)





Read About Doug Engelbart at…



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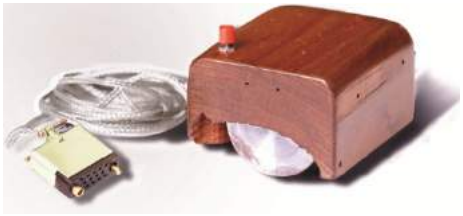


Click here

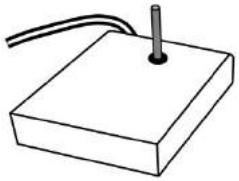
http://www.dougengelbart.org/

HCI's First User Study[1]

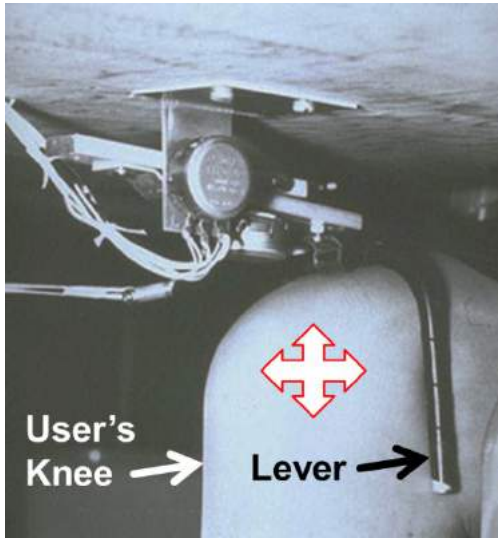
A comparative evaluation of…



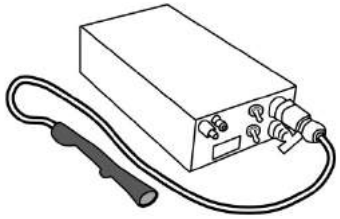
Mouse



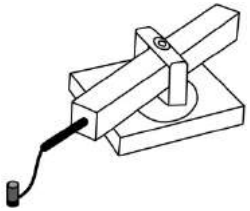
Joystick



Knee-controlled lever



Lightpen



Grafacon

[1] English, W. K., Engelbart, D. C., & Berman, M. L. (1967). Display selection techniques for text manipulation. *IEEE Transactions on Human Factors in Electronics*, *HFE-8*(1), 5-15.

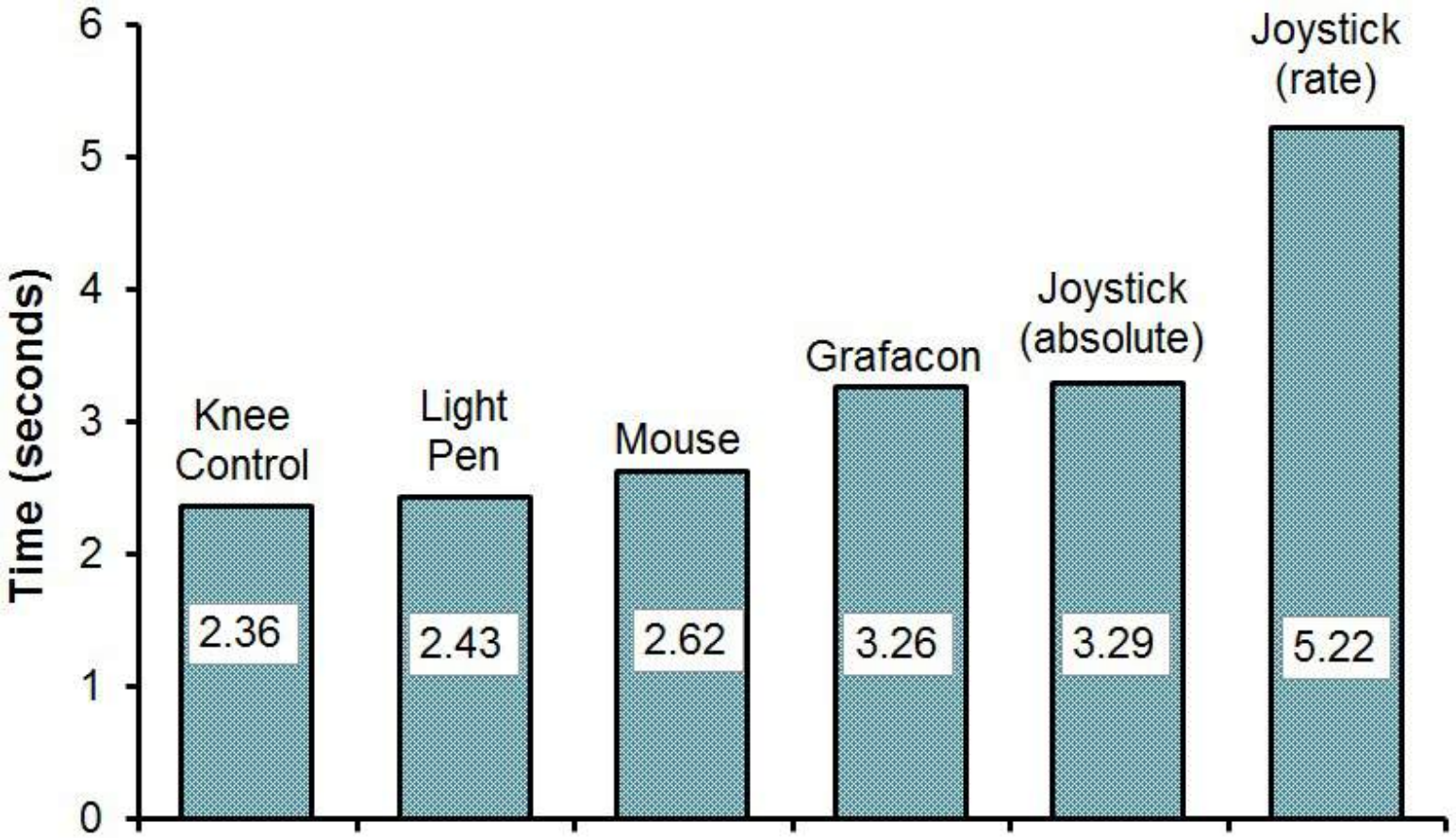
Click here

http://ieeexplore.ieee.org

Experiment Design

- Participants: 13
- **Independent variable (settings)**
  - "Input method" with six levels: mouse, light pen, Grafacon, joystick (position-control), joystick (rate-control), knee-controlled lever
- **Dependent variables (metrics)**
  - Task completion time, error rate
  - (Note: task completion time = access time + motion time)
- Within-subjects, counterbalanced
- Task:
  - Press spacebar, acquire device, position cursor on target, select target

Results (1)

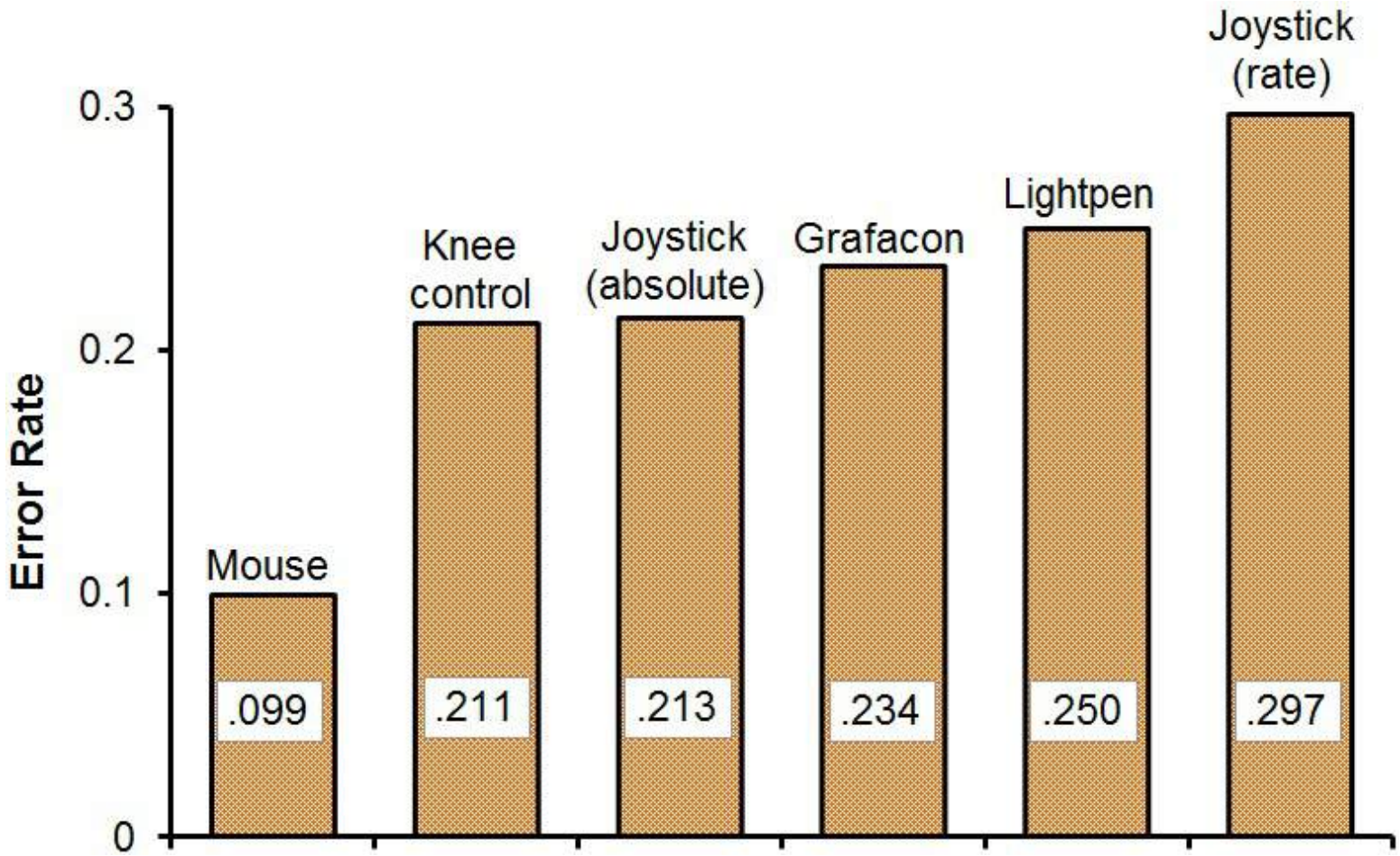


Notes:

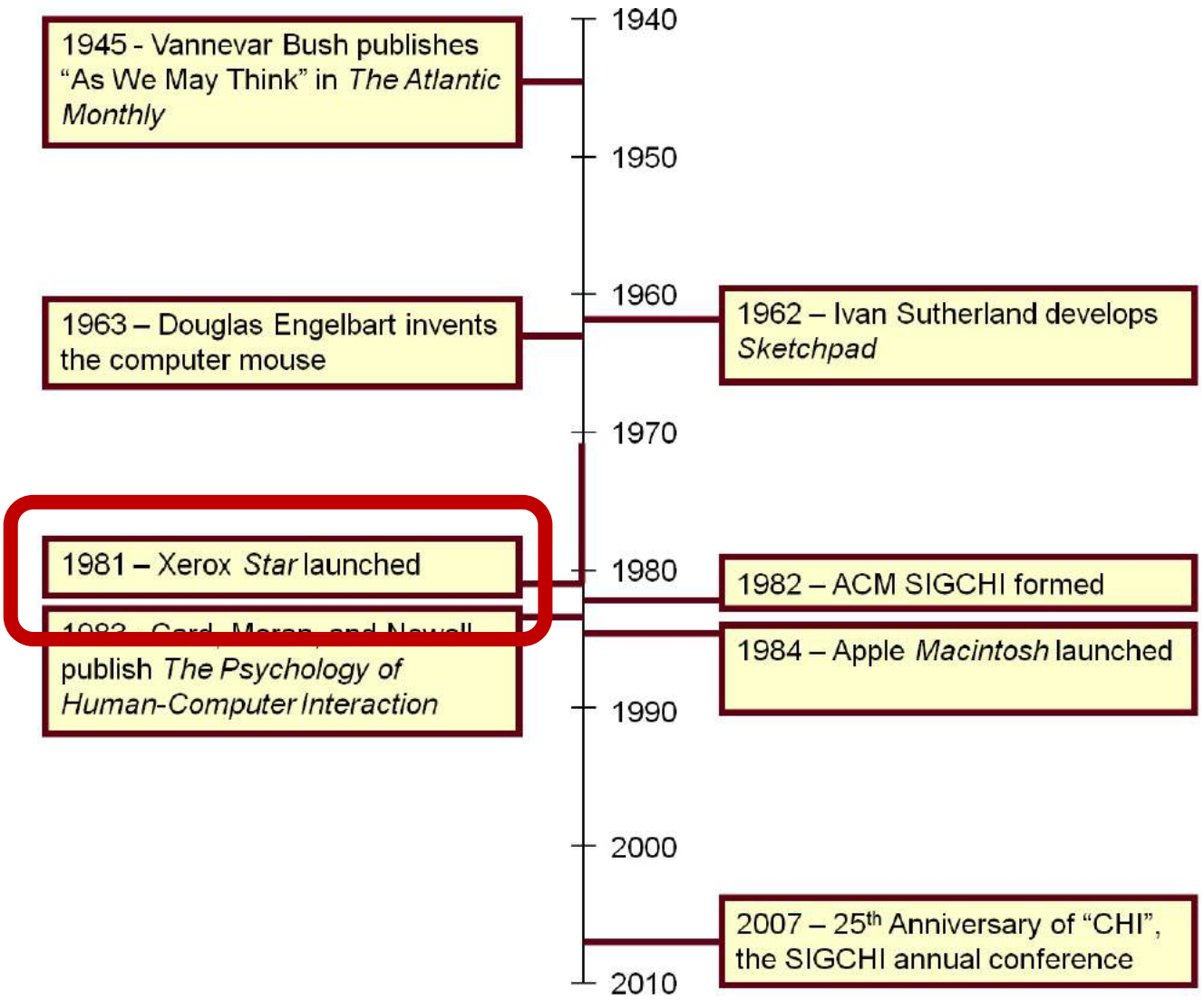
[1] Access time with the knee-controlled lever was zero (since the device is always "acquired").

[2] Light pen use is fatiguing, since the user's arm is held in the air in front of the display.

Results (2)

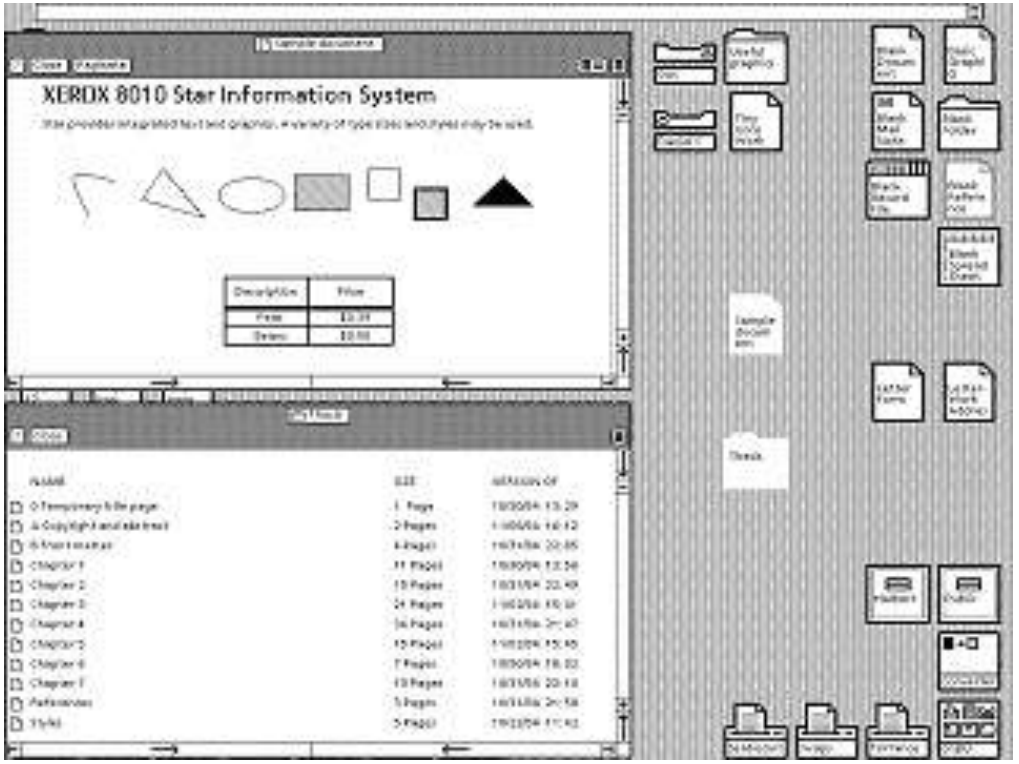

Error Rate
0.3
0.2
0.1
0
Mouse
.099
Knee control
.211
Joystick (absolute)
.213
Grafacon
.234
Lightpen
.250
Joystick (rate)
.297

Timeline of Significant Events



Xerox *Star* (1981)

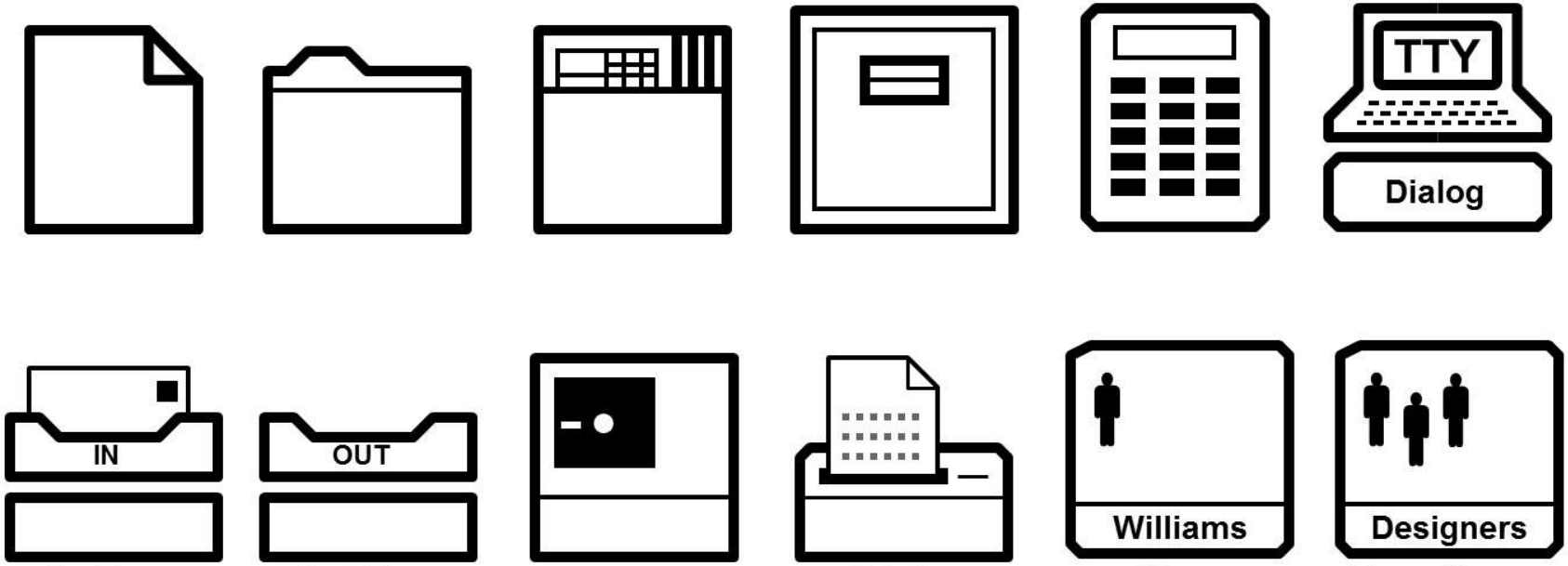




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Steve Jobs Introducing First Macintosh

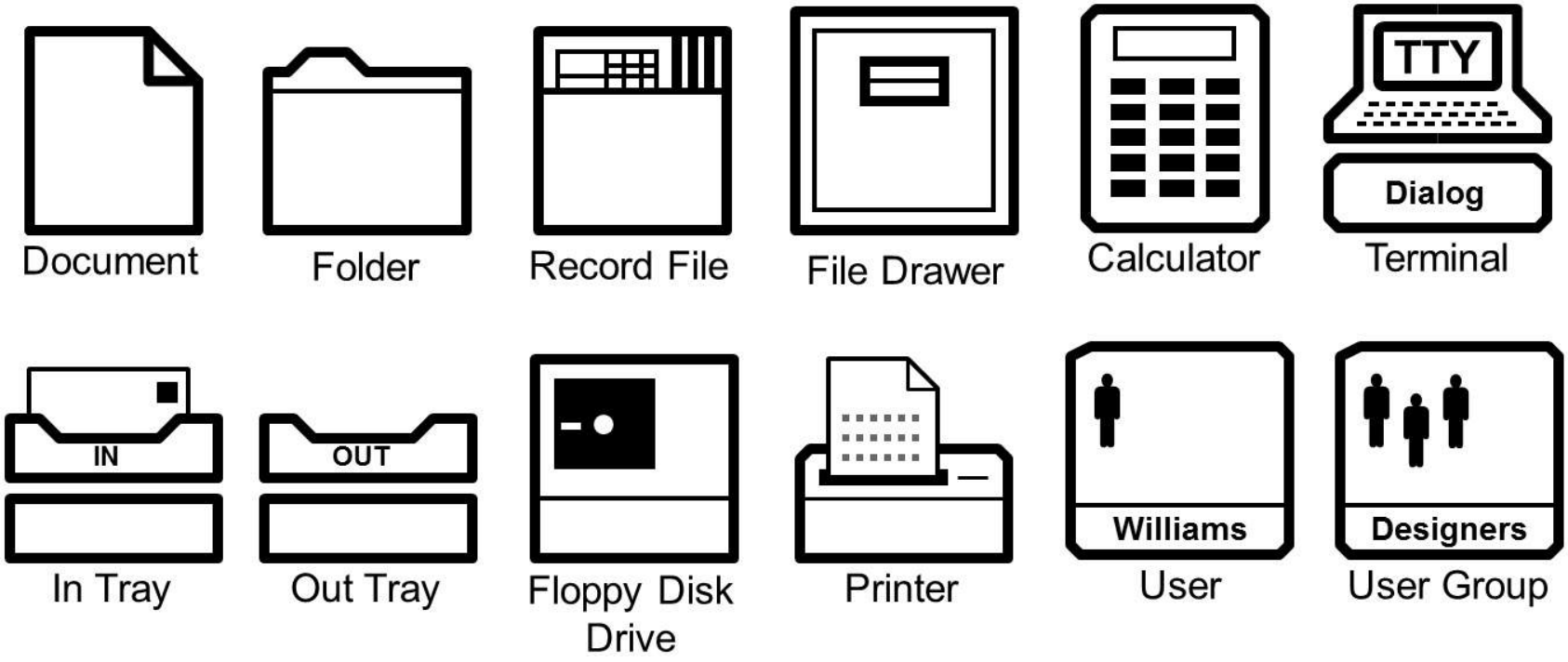
Research Topic – Icon Recognition



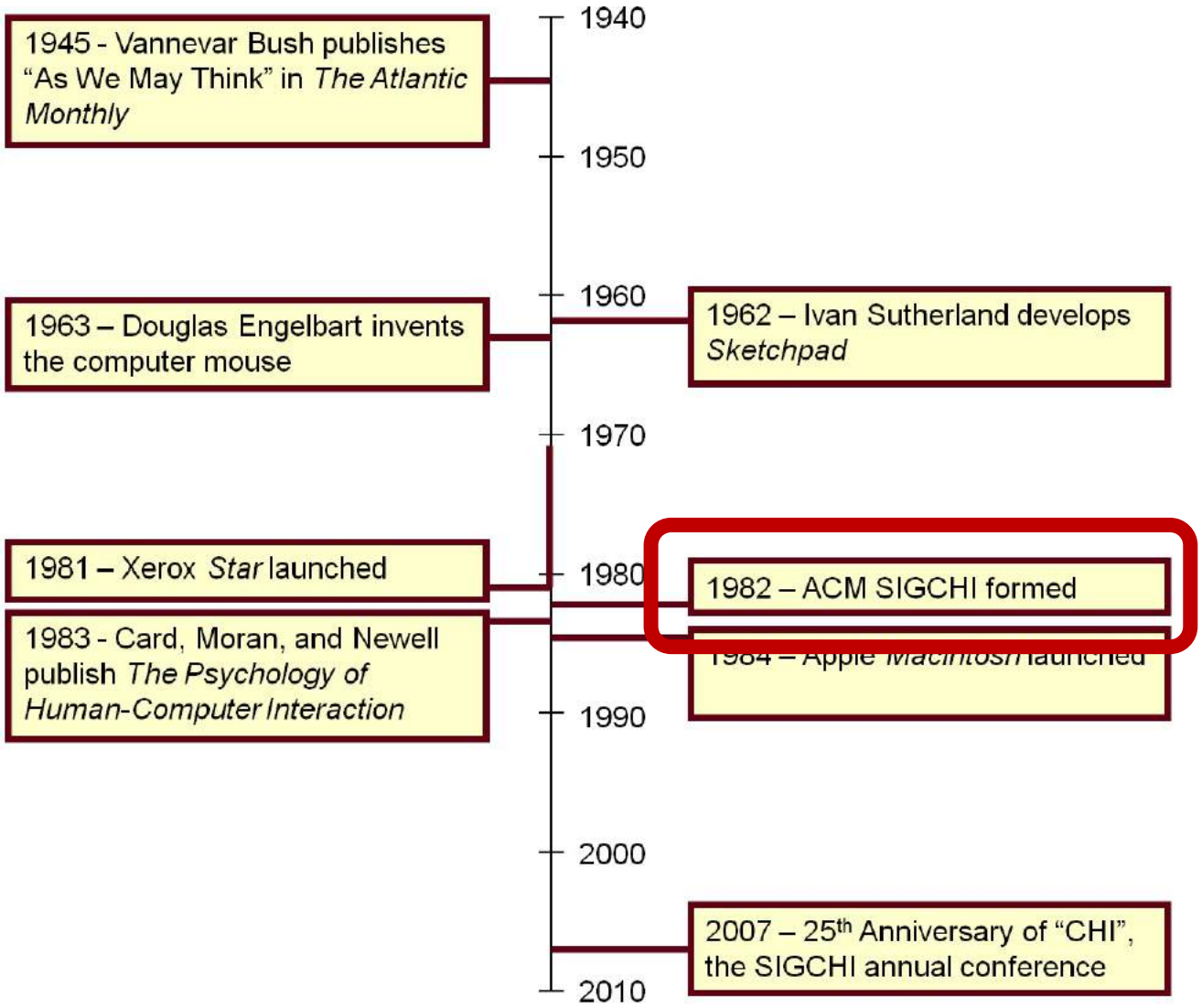
Can you guess the purpose of each icon?

(See also Figure 3.19 and Student Exercise 3.1.)

Star GUI Icons



Timeline of Significant Events



Birth of HCI - 1983

- Notable events:
  1. First ACM SIGCHI conference (1983)
  2. Publication of *The Psychology of Human-Computer Interaction* by Card, Moran, and Newell (1983)
  3. Apple *Macintosh* announced via brochures (December, 1983) and launch (January, 1984)

ACM SIGCHI Mission

The ACM Special Interest Group on Computer-Human Interaction is the world's largest association of professionals who work in the research and practice of **computer-human interaction**. This interdisciplinary group is composed of **computer scientists, software engineers, psychologists, interaction designers, graphic designers, sociologists, and anthropologists**, just to name some of the domains whose special expertise come to bear in this area. They are brought together by a shared understanding that **designing useful and usable technology** is an interdisciplinary process, and believe that when done properly it has the power to **transform persons' lives**.

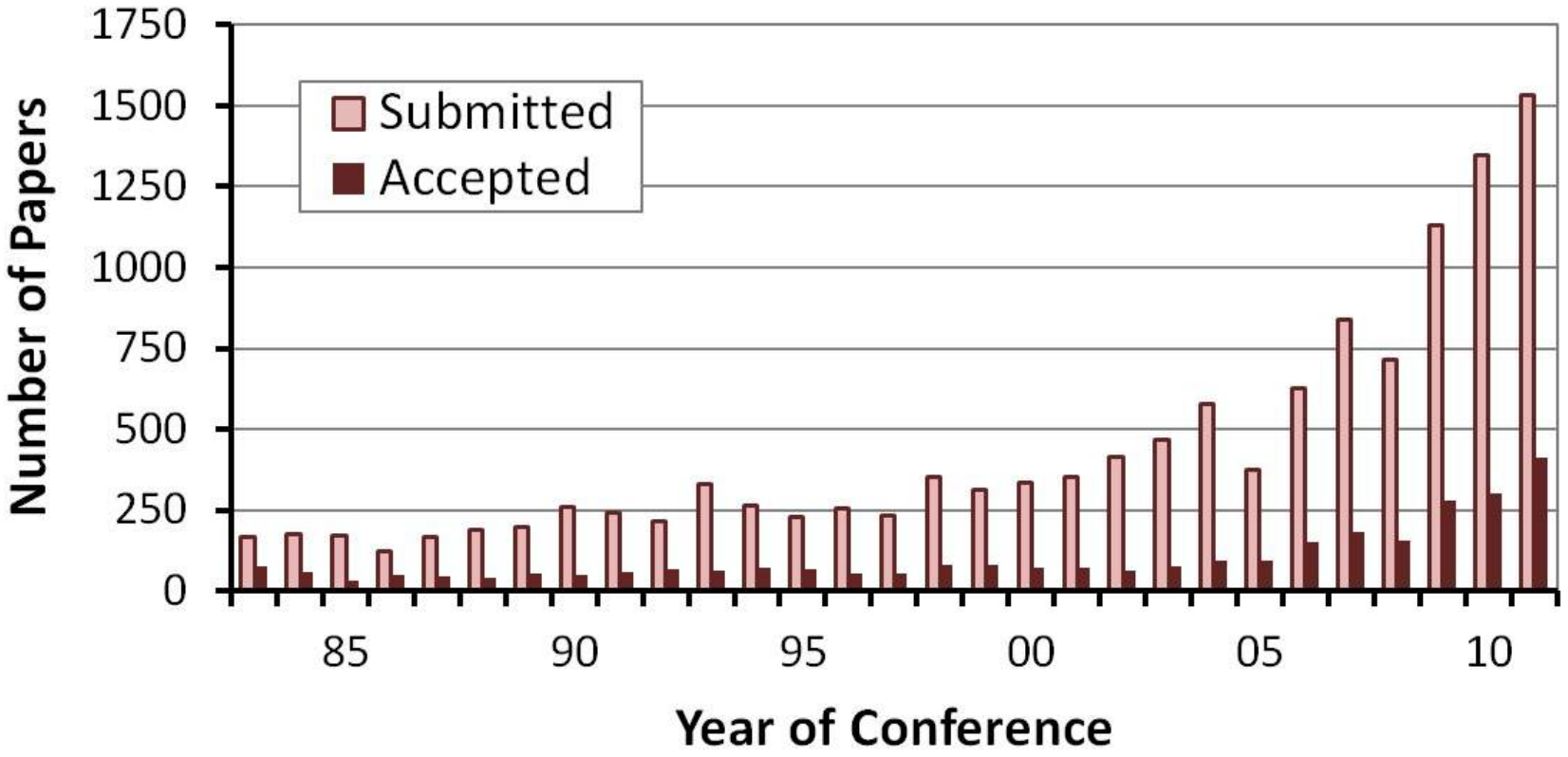
SIGCHI Web Site



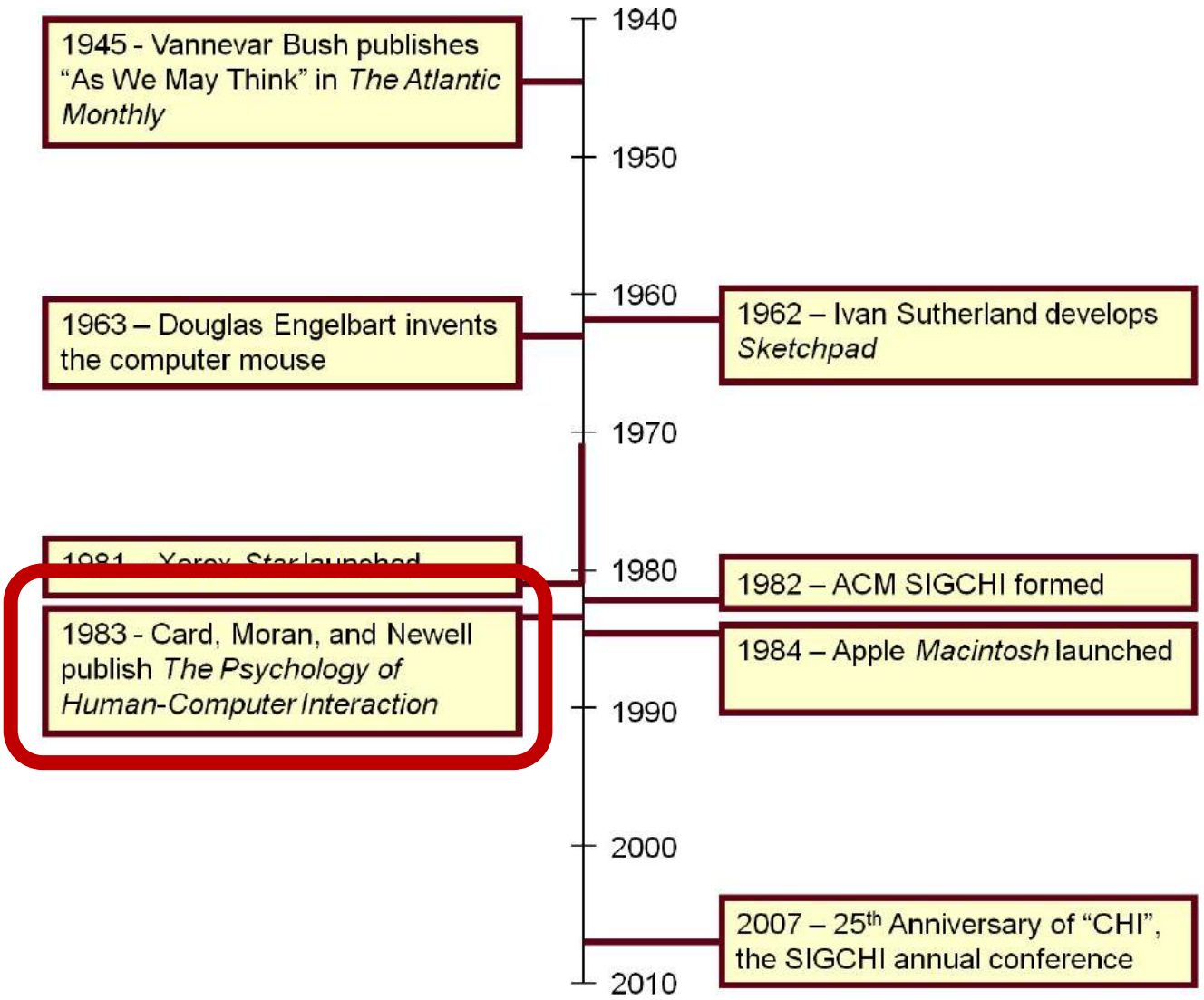
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http://www.sigchi.org/

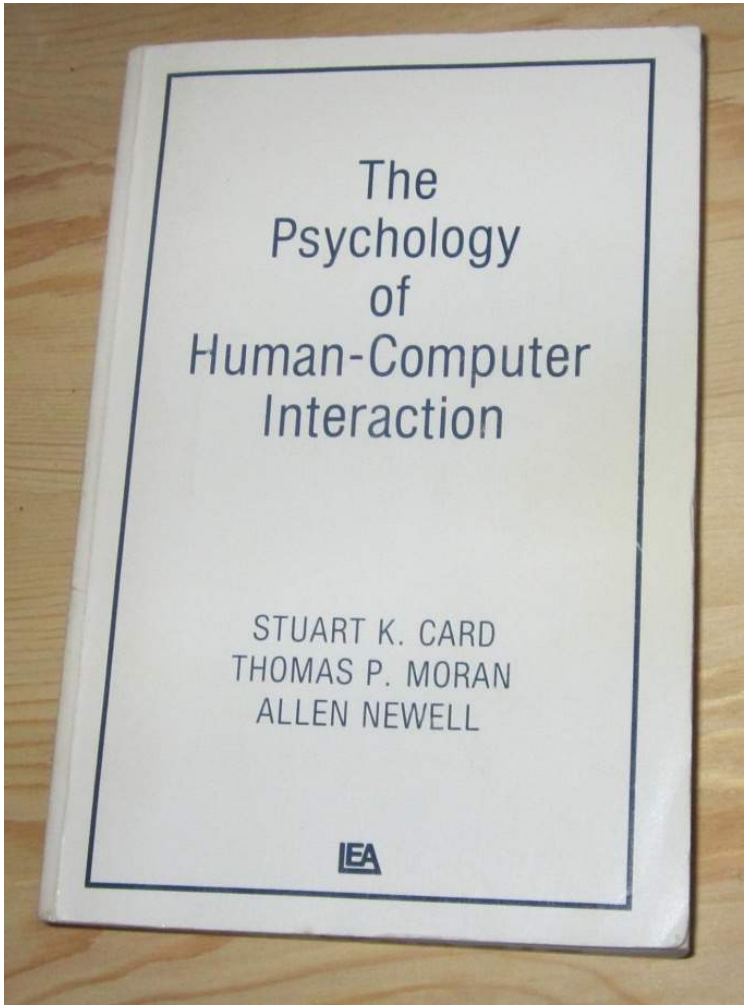
SIGCHI Conference Publications



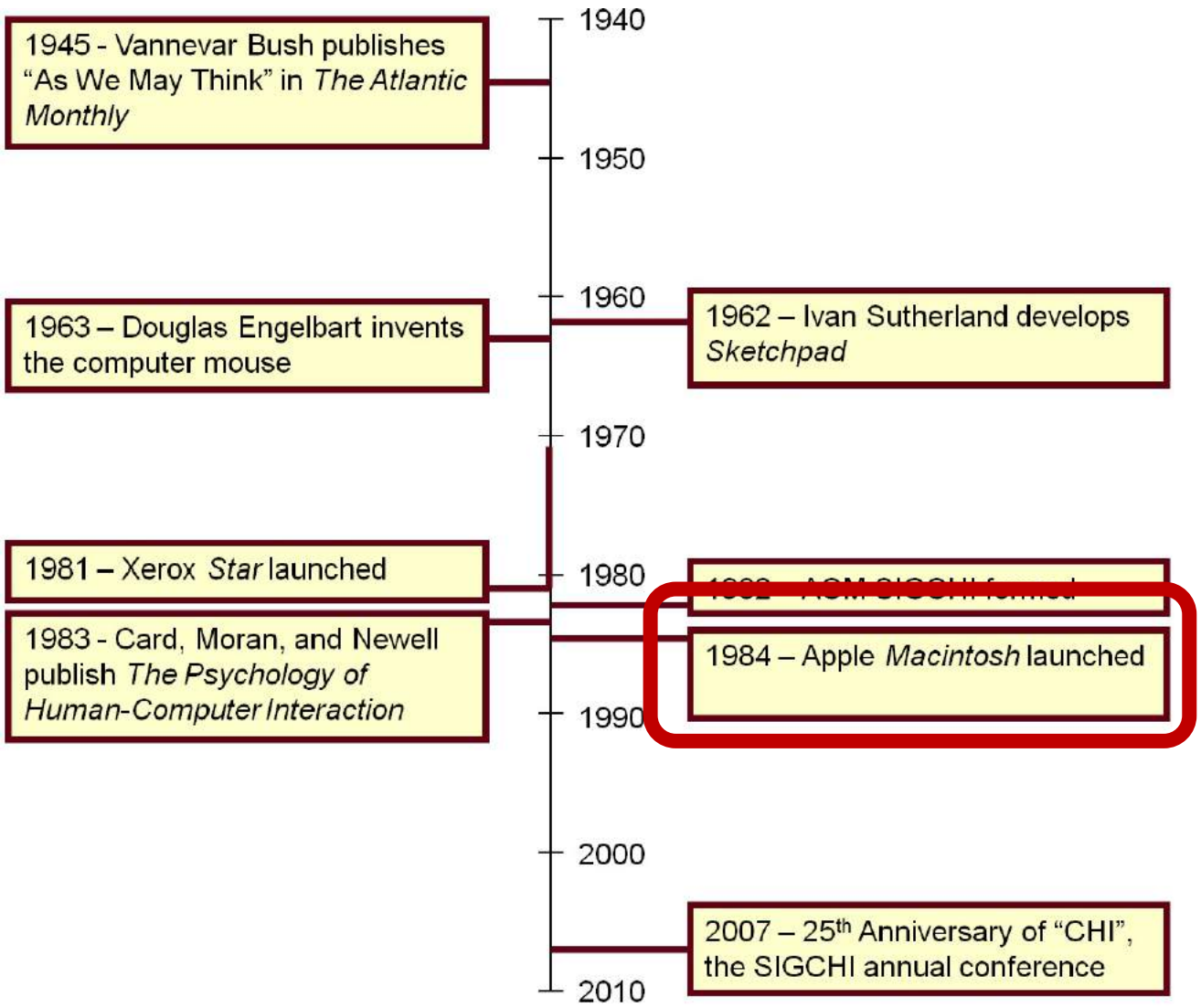
Timeline of Significant Events



The Psychology of Human-Computer Interaction
Card, Moran, and Newell (1983)



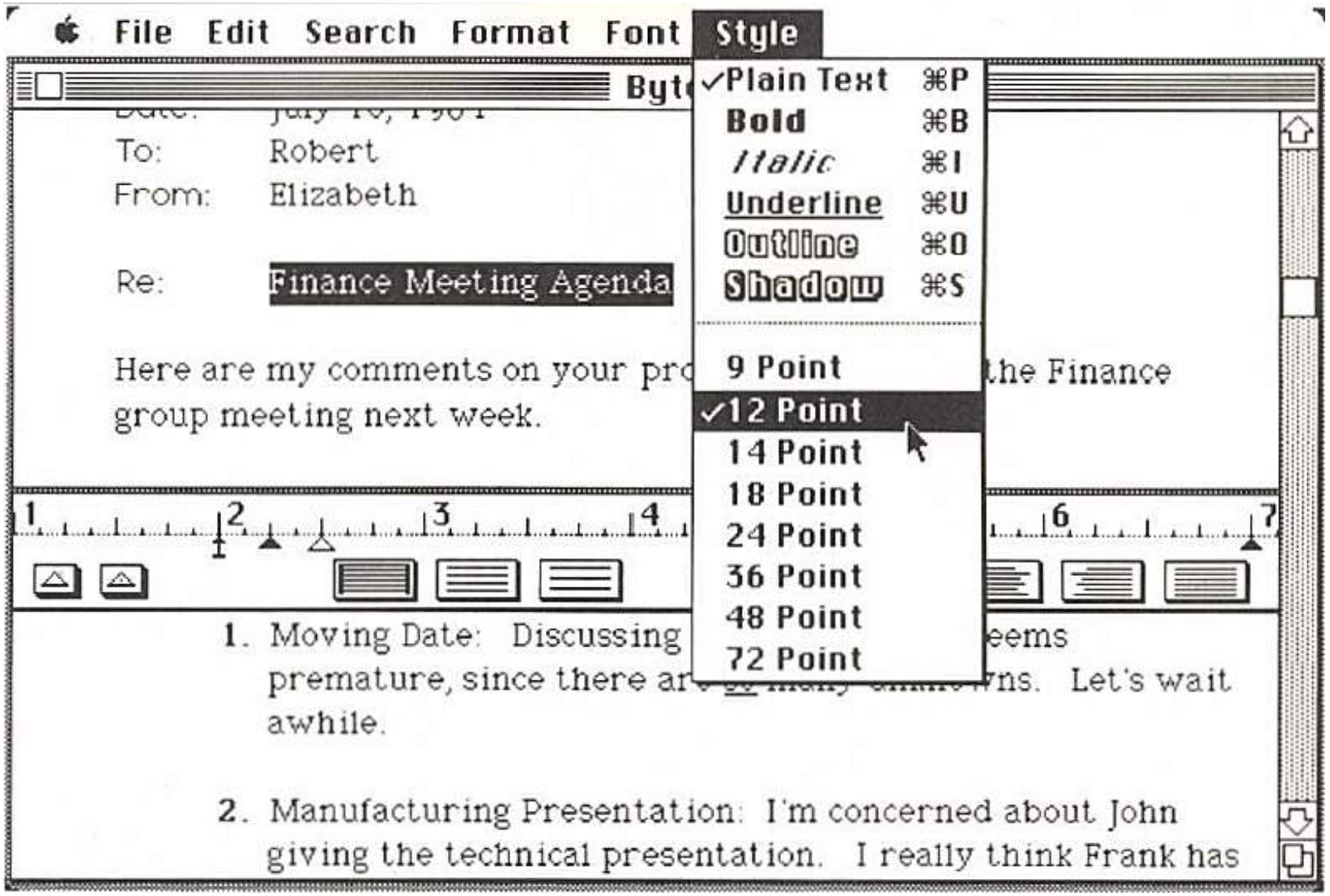
Timeline of Significant Events



Apple *Macintosh* (1984)



MacWrite Software



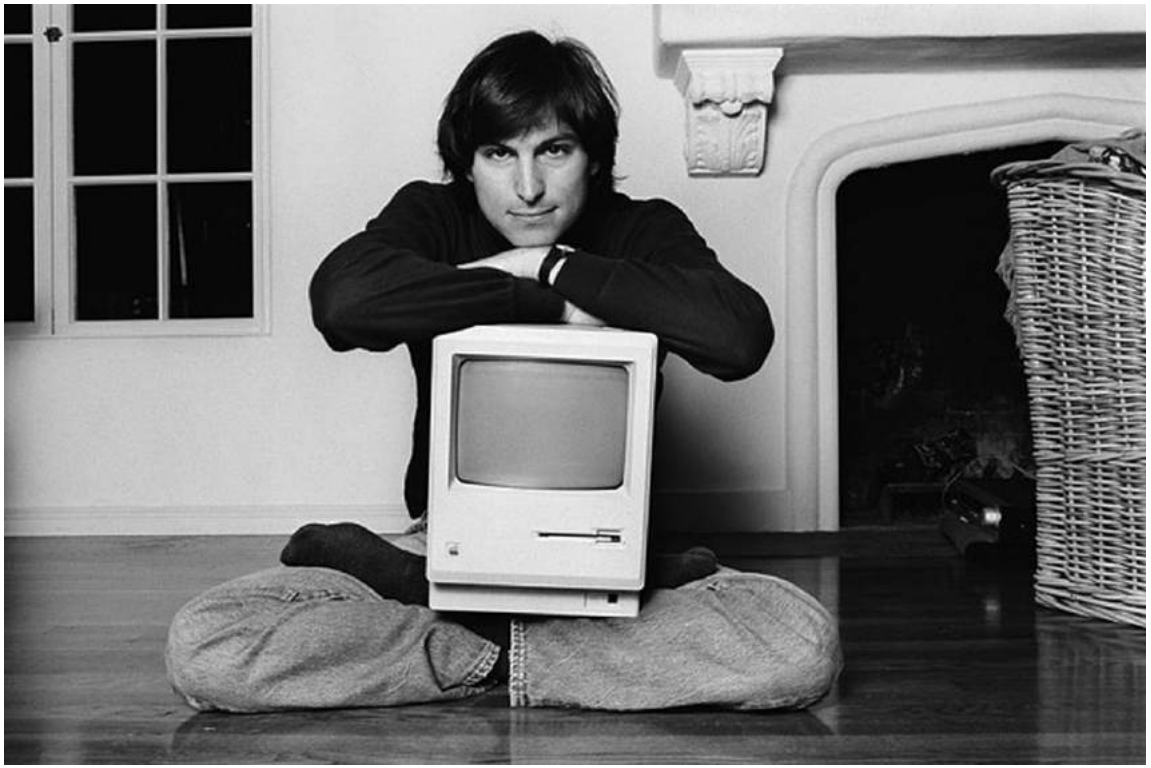
Apple *Macintosh* Commercial (1984)



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https://www.youtube.com/results?search_query=apple+macintosh+commercial+1984

Apple *Macintosh* Commercial (1984)



Click here

Apple *Macintosh* Timeline

Click here

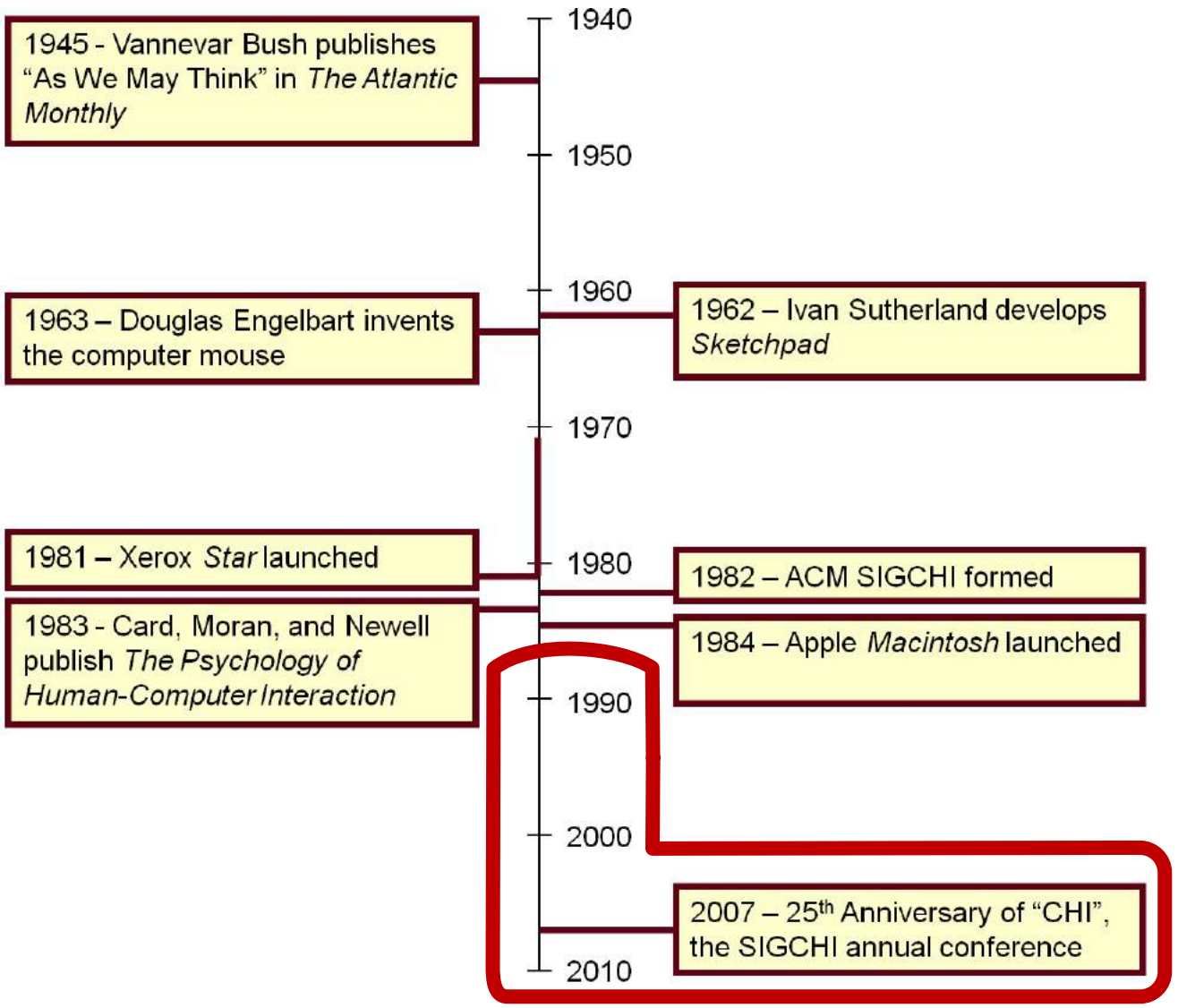
1976	April – Apple Computer Inc. founded in Cupertino, California.
1977	Launch of Apple II. Sells for $1300 U.S. with 4KB RAM. Hugely successful (more than one million units sold). Works with a text-based command-line interface.
1978	*Lisa* project started . Goal of producing a powerful (and expensive!) personal computer.
1979	September – *Macintosh* project started. Goal of producing a low-cost easy-to-use computer for the average consumer. December – Apple and Xerox sign an agreement that allows Xerox to invest in Apple. In return Apple's engineers visit Xerox PARC and see the Xerox *Alto*. The GUI ideas in the *Alto* influence *Lisa* and *Macintosh* development.
1980	December – Apple goes public through initial public offering (IPO) of its stock.
1981	May – Xerox *Star* launched at the National Computer Conference (NCC) in Chicago. Members of the *Lisa* design team are present and see the *Star* demo. They decide to re-vamp the *Lisa* interface to be icon-based. August – IBM PC announced. Highly successful, but embodies traditional text-based command-line interface.
1982	*Lisa* and *Macintosh* development continue. Within Apple, there is an atmosphere of competition between the two projects
1983	January – *Lisa* released. *Lisa* incorporates a GUI and mouse input. Sells for $10,000 U.S. In the end, *Lisa* is a commercial failure. December -- brochures distributed in magazines (e.g., *Time*) pre-announcing the *Macintosh.*
1984	January 22 – *Macintosh* ad plays during Super Bowl XVIII. January 24 – *Macintosh* released. Sells for $2500 U.S.

Microsoft *Windows* History



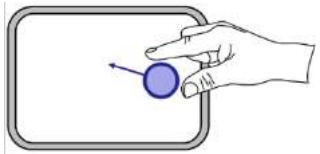
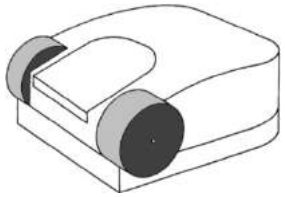
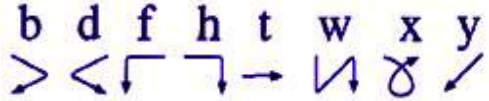
Click here

Timeline of Significant Events



HCI Research

- Research precedes products
- Consider…
  - Two-finger gestures (Apple *iPhone*, 2007)
  - Acceleration-sensing (Nintendo *Wiimote*, 2005)
  - Wheel mouse (Microsoft *Intellimouse*, 1996)
  - Single-stroke text input (Palm's *Graffiti*, 1995)
- Were these ideas born out of engineering or design brilliance? Not really…

- Two-finger gestures: ~~2007?~~  **1978** [1]
- Acceleration-sensing: ~~2005?~~  **1998** [2]
- Wheel mouse: ~~1996?~~  **1993** [3]
- Single-stroke text input: ~~1995?~~ 

1993 [4]

[1] Herot, C. F., & Weinzapfel, G. (1978). One-point touch input of vector information for computer displays. *Proc SIGGRAPH '78*, 210-216, New York: ACM.
[2] Harrison, B., Fishkin, K. P., Gujar, A., Mochon, C., & Want, R. (1998). Squeeze me, hold me, tilt me! An exploration of manipulative user interfaces. *Proc CHI '98*, 17-24, New York: ACM.
[3] Venolia, D. (1993). Facile 3D manipulation. *Proc CHI '93*, 31-36, New York: ACM.
[4] Goldberg, D., & Richardson, C. (1993). Touch-typing with a stylus. *Proc CHI '93*, 80-87, New York: ACM.

Resources

Google Scholar:	`http://scholar.google.ca/`
ACM Digital Library:	`http://portal.acm.org/`
HCI Bibliography:	`http://hcibib.org/`
Wikipedia:	`http://en.wikipedia.org/`
Book web site:	`http://www.yorku.ca/mack/HCIbook`

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